
Assignment on Performance & Load Testing using Apache JMeter

Submission Deadline: 18 September, 2026 — 11:59 pm

1. Introduction & Objectives

Apache JMeter is an open-source, Java-based software application designed to perform load testing, stress testing, and performance measurement of web applications and service protocols. By simulating heavy concurrent user workloads, JMeter enables software engineers to evaluate application stability, identify architectural performance bottlenecks, and measure key performance metrics such as response time, throughput, and error rates.

The objective of this assignment is to gain hands-on experience in configuring automated performance tests, recording HTTP request samples, applying assertions to validate response thresholds, and conducting quantitative performance analysis across varying user loads.

2. Target Applications & Request Types

You are required to perform load testing on the **LoadLab** application hosted at <http://103.94.135.91:8080>. To ensure testing across a diverse set of HTTP request methods, payload sizes, and server behaviors, your test plan must include all **five distinct target endpoints** described below:

1. **Home Page (GET /):** Baseline web page request (0–30 ms latency, 2 KB response body).
2. **Notice Board (GET /notices):** Dynamic page request with random uniform delay (100–400 ms latency, 7 KB response body).
3. **Course Catalogue (GET /courses):** CPU-intensive request with latency that scales under user load (150–250 ms + CPU work, 43 KB response body).
4. **User Authentication (POST /login):** Form-urlencoded POST request passing credentials (`username=student`, `password=student123`) to evaluate session cookie setting and 303 redirection behavior.
5. **Bandwidth Data Transfer (GET /api/download/256 or POST /api/upload):** High-throughput API endpoint designed specifically to measure data reception and transmission rates (KB/s).

Note: The target server operates on plain HTTP (no SSL/TLS certificate setup required in browser). Because the environment is shared across the entire class, do not exceed 100 threads or run infinite loop iterations on high-bandwidth endpoints.

3. Test Requirements & Setup Specifications

Your test plan must adhere strictly to the following parameters and configurations:

- **Load Profiles (Thread Groups):** Execute separate test runs for two distinct workload conditions:
 - a) **Low Load Profile:** 50 concurrent samples (threads).

-
- b) **High Load Profile:** 100 concurrent samples (threads).
 - **Ramp-Up Period:** Configure the Ramp-Up Period to **100 seconds** for all test executions.
 - **Logic Controllers:** Organize HTTP request samples using separate **Recording Controllers** for each of the target endpoints.
 - **Test Scripting:** Incorporate an **HTTP(S) Test Script Recorder** under *Non-Test Elements* to intercept real browser interactions and automatically generate request samples.
 - **Duration Assertion:** Apply a **Duration Assertion** across the HTTP request samples.

Assertion Threshold Constraint: Set the maximum allowed duration threshold (in milliseconds) such that **both successful and failed events are observed** during test execution (e.g., if the baseline average response time is 250 ms, set the duration assertion to 200 ms to trigger controlled assertion failures).

- **Listeners & Configuration:** Attach necessary listeners to your test plan, including *View Results Tree*, *Aggregate Report*, and *Summary Report*, to capture live performance metrics and output log data.

4. Deliverables

You must submit a single compressed zip file containing the required project artifacts along with a public/unlisted YouTube video demonstration link:

1. `yourID.Assessment.docx` (or `yourID.Assessment.pdf`): A comprehensive performance evaluation report detailing:
 - a) **Execution Time Analysis:** A comparative summary table of Shortest (Minimum), Longest (Maximum), and Average execution times (ms) for 50 vs. 100 samples across all 5 target endpoints.
 - b) **Throughput & Bandwidth Analysis:** A comparative evaluation of Average Throughput (requests/sec), Average Data Reception Rate (KB/sec), and Average Data Transmission Rate (KB/sec) for 50 vs. 100 samples.
 - c) **Assertion & Error Rate Analysis:** Discussion on the observed impact of the Duration Assertion on overall success/failure rates.
 - d) **Video Demonstration Link:** Prominently feature the YouTube link to your video demonstration at the beginning of your report document.
2. `yourID.testPlan.jmx`: The exported JMeter Test Plan file containing all Thread Groups, Recording Controllers, Assertions, and Listener configurations.
3. `yourID.result.csv`: Raw aggregate performance test results exported directly from JMeter execution logs.
4. `yourID.html.zip`: Compressed archive of the automatically generated JMeter HTML Dashboard Report containing performance charts and graphs.
5. **YouTube Video Demonstration (Mandatory):** You must record a clear screen video of your entire assignment workflow. **Crucial Requirement:** You must be clearly visible via webcam picture-in-picture (similar to a Zoom recording showing both speaker and screen contents) throughout the demonstration. The video must show:

- Webcam overlay showing yourself explaining the workflow live.
- Step-by-step JMeter test plan and proxy/recording setup.
- Test execution runs for both 50 and 100 thread profiles.
- Observation of Duration Assertion passes/failures and HTML report generation.

Upload this video to YouTube (Public or Unlisted) and provide the working link inside your report document.

5. Summary Report Template

Structure the performance metric tables in your report according to the format shown below:

Table 1: Execution Time and Throughput Comparison Across 5 Request Endpoints (50 vs. 100 Samples)

Target Web Endpoint	Samples	Min (ms)	Max (ms)	Avg (ms)	Throughput	Error %
Home Page (GET /)	50	—	—	—	— req/s	— %
	100	—	—	—	— req/s	— %
Notice Board (GET /notices)	50	—	—	—	— req/s	— %
	100	—	—	—	— req/s	— %
Course Catalogue (GET /courses)	50	—	—	—	— req/s	— %
	100	—	—	—	— req/s	— %
User Login (POST /login)	50	—	—	—	— req/s	— %
	100	—	—	—	— req/s	— %
Data API (GET /api/download/256)	50	—	—	—	— req/s	— %
	100	—	—	—	— req/s	— %

6. Evaluation Rubric

Evaluation Criteria	Description	Weight
Video Demonstration	Complete screen recording showing test setup, execution, dashboard generation, and student (visible in picture-in-picture)	60 %
Test Plan Design	Proper configuration of HTTP(S) Script Recorder, Recording Controllers, and Thread Groups	15 %
Duration Assertion	Correct threshold setup displaying both passed and failed assertion events	10 %
Test Artifacts	Complete JMX script, raw CSV results, and generated HTML dashboard report	10 %
Analysis & Report	Quality of comparative tables, execution time metrics, throughput, and bandwidth analysis	5 %